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 Studierichting: Informatica
 Oefeningenreeks 3G nr. 11.1

$$f(x) = \frac{2x}{x^2 + 1}$$

domein f : $\mathbb{R}$

nulpunten $f(x) = 0 \Leftrightarrow x = 0$

$$\begin{array}{c|c} x & 0 \\ \hline f(x) & - \quad 0 \quad + \end{array}$$

Geen VA

$$\lim_{x \rightarrow \infty} \frac{2x}{x^2} = \lim_{\infty} \frac{2}{x} = 0$$

$$\begin{array}{c|c} x & 0 \\ \hline f(x) - A & - \quad | \quad + \end{array}$$

Er is een HA, dus er is geen SA.

$$f'(x) = \frac{(x^2 + 1) \cdot 2 - 2x \cdot 2x}{(x^2 + 1)^2} = \frac{-2x^2 + 2}{(x^2 + 1)^2}$$

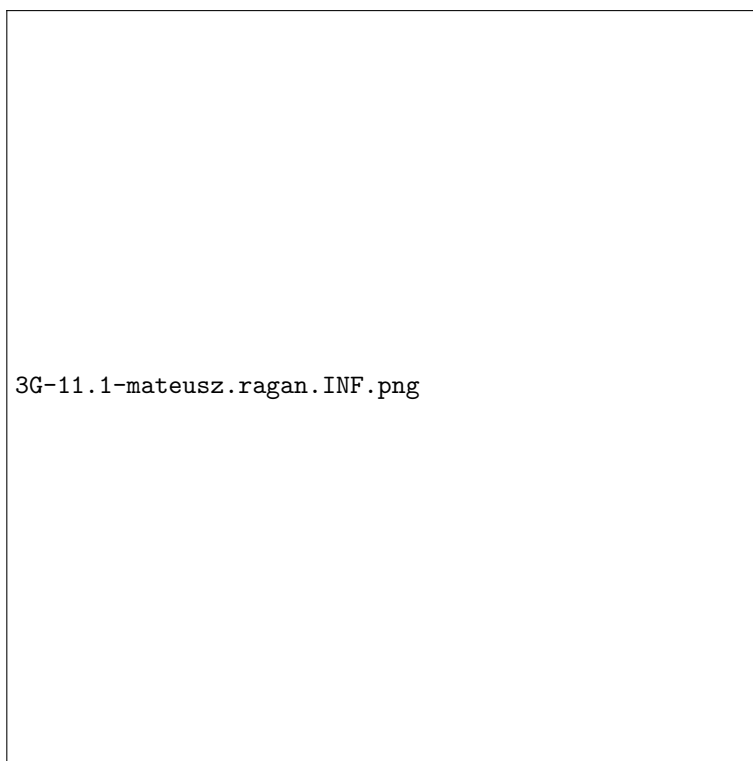
$$\begin{array}{c|c} x & -1 \quad \quad 1 \\ \hline f'(x) & - \quad 0 \quad + \quad 0 \quad - \\ f(x) & \searrow \quad \min(-1) \quad \nearrow \quad \max(1) \quad \searrow \end{array}$$

$$f''(x) = \frac{(x^2 + 1)^2 \cdot -4x - ((-2x^2 + 2)(4x^3 + 4x))}{(x^2 + 1)^4} = \frac{4x^5 - 8x^3 - 12x}{(x^2 + 1)^4}$$

$$= \frac{4x(x^4 - 2x^2 - 3)}{(x^2 + 1)^4} = \frac{4x(x^2 + 1)(x^2 - 3)}{(x^2 + 1)^4} = \frac{4x(x^2 - 3)}{(x^2 + 1)^3}$$

$$\begin{array}{c|c} x & -\sqrt{3} \quad \quad 0 \quad \quad \sqrt{3} \\ \hline f''(x) & - \quad 0 \quad + \quad 0 \quad - \quad 0 \quad + \\ f(x) & \frown \quad \text{bgpt}\left(\frac{-\sqrt{3}}{2}\right) \quad \smile \quad \text{bgpt}(0) \quad \frown \quad \text{bgpt}\left(\frac{\sqrt{3}}{2}\right) \quad \smile \end{array}$$

References



3G-11.1-mateusz.ragan.INF.png